

08. UI Wireframes

8.1 Introduction

The User Interface (UI) Wireframes represent the initial visual design of the Lab Resource Utilization Platform. These wireframes illustrate the layout of different application screens, helping stakeholders understand the navigation flow, user interactions, and overall system usability before implementation.

The wireframes are designed with simplicity, consistency, and ease of use in mind, ensuring that users from different institutions can efficiently manage laboratory resources.

8.2 Login Screen

1. Login Page (UI Wireframe)

LRUP
Lab Resource Utilization Platform

Welcome Back!
Login to your account and manage laboratory resources, bookings, maintenance and more.

Login to Your Account
Enter your credentials to continue

Email
Enter your email address

Password
Enter your password

☐ Remember me [Forgot Password?](#)

[Login](#)

OR

[Login with SSO](#)

Don't have an account? [Register here](#)

Secure Access

The Login page allows registered users to securely access the system using their email and password. User authentication is handled using JWT-based authentication with role-based access control.

8.3 Registration Screen

2. Registration Page (UI Wireframe)

LRUP
Lab Resource Utilization Platform

Create Your Account

Register to access and manage laboratory resources, bookings, maintenance and more.

Create Account
Fill in the details to register

First Name *
Enter first name

Last Name *
Enter last name

Email *
Enter email address

Phone
Enter phone number

Institution *
Select institution

Department
Select department

Role *
Select role

Password *
Enter password

Confirm Password *
Confirm password

[Register](#)

OR

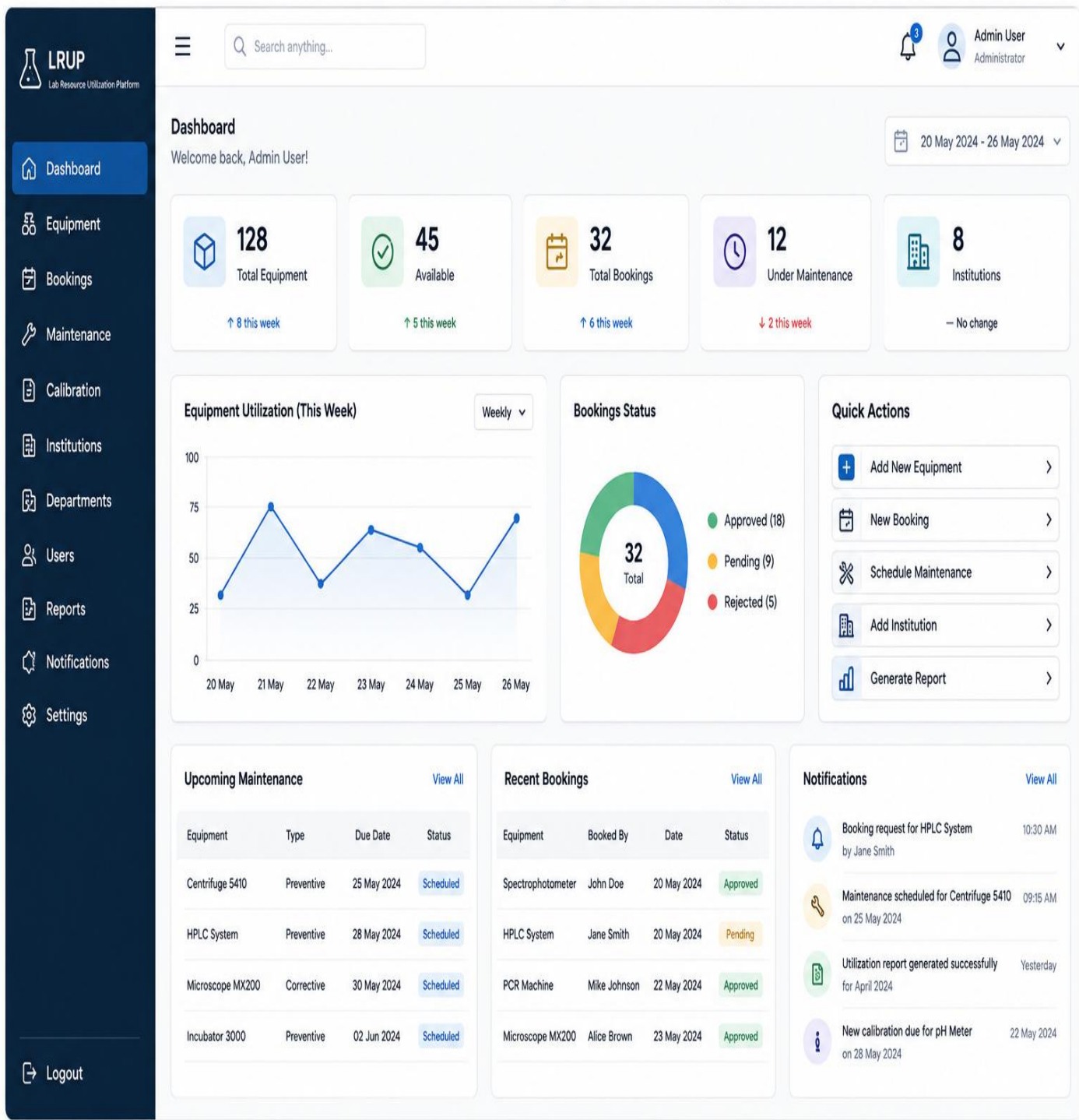
[Sign up with Google](#)

Already have an account? [Login here](#)

The Registration page enables new users to create an account by providing personal information, institution details, and role selection

8.4 Dashboard

3. Dashboard (UI Wireframe)



The Dashboard provides a summarized overview of equipment availability, booking statistics, maintenance schedules, utilization reports, and system notifications.

8.5 Equipment Management



LRUP
Lab Resource
Utilization Platform

Dashboard

Equipment

Bookings

Maintenance

Calibration

Institutions

Departments

Users

Reports

Notifications

Settings

Logout

Search anything...

4



Admin User
Administrator

Equipment Management

Manage and monitor all laboratory equipment

+ Add New Equipment

128
Total Equipment

45
Available

32
Under Maintenance

8
Out of Service

12
Departments

Search equipment...

All Categories

All Departments

All Status

Reset

#	Equipment Name	Category	Department	Status	Location	Actions
1	Centrifuge 5410	Centrifuges	Biotechnology	Available	Lab 1, Block A	  
2	HPLC System	Chromatography	Chemistry	In Use	Lab 2, Block B	  
3	Microscope MX200	Microscopes	Biotechnology	Available	Lab 1, Block A	  
4	Incubator 3000	Incubators	Microbiology	Maintenance	Lab 3, Block C	  
5	Spectrophotometer	Spectroscopy	Chemistry	Available	Lab 2, Block B	  
6	PCR Machine	Molecular Biology	Biotechnology	In Use	Lab 3, Block C	  
7	pH Meter	Measurement	Chemistry	Out of Service	Lab 2, Block B	  
8	Hot Air Oven	Ovens	Microbiology	Available	Lab 1, Block A	  

Showing 1 to 8 of 128 entries

<

1

2

3

4

5

...

16

>

This screen enables laboratory administrators to add, update, delete, and monitor equipment. Equipment status, department allocation, and maintenance history are also displayed.

8.6 Booking Management

5. Booking Management (UI Wireframe)



LRUP

Lab Resource Utilization Platform

Dashboard

Equipment

Bookings

Maintenance

Calibration

Institutions

Departments

Users

Reports

Notifications

Settings

Logout

Search anything...

3



Admin User
Administrator

Booking Management

Manage equipment bookings and requests

+ New Booking

32

Total Bookings

18

Approved

9

Pending

5

Rejected

12

This Month

Search bookings...

All Status

All Departments

20 May 2024 - 26 May 2024

Reset

#	Booking ID	Equipment	Booked By	Department	From Date	To Date	Purpose	Status	Actions
1	BK-2024-001	HPLC System	Jane Smith	Chemistry	20 May 2024	21 May 2024	Research Work	Approved	   
2	BK-2024-002	Microscope MX200	Mike Johnson	Biotechnology	21 May 2024	22 May 2024	Sample Analysis	Pending	   
3	BK-2024-003	Centrifuge 5410	Alice Brown	Microbiology	22 May 2024	22 May 2024	Routine Test	Approved	   
4	BK-2024-004	PCR Machine	Robert Wilson	Biotechnology	23 May 2024	24 May 2024	PCR Amplification	Approved	   
5	BK-2024-005	Spectrophotometer	Emily Davis	Chemistry	24 May 2024	24 May 2024	Concentration Check	Rejected	   
6	BK-2024-006	Incubator 3000	Daniel Lee	Microbiology	25 May 2024	26 May 2024	Culture Incubation	Pending	   
7	BK-2024-007	pH Meter	Olivia Martin	Chemistry	26 May 2024	26 May 2024	pH Measurement	Approved	   
8	BK-2024-008	Hot Air Oven	Sophia Taylor	Microbiology	27 May 2024	28 May 2024	Sterilization	Pending	   
9	BK-2024-009	HPLC System	James Anderson	Chemistry	28 May 2024	29 May 2024	Data Collection	Approved	   
10	BK-2024-010	Microscope MX200	William Thomas	Biotechnology	29 May 2024	29 May 2024	Cell Observation	Rejected	   

Showing 1 to 10 of 32 entries

<

1

2

3

4

>

The Booking screen allows users to search available equipment, create booking requests, and track booking approval status.

8.7 Institution & Department Management

8. Institution Management (UI Wireframe)

LRUP
 Lab Resource
 Utilization Platform

Admin User Administrator

Institution Management

Manage institutions and their details

+ Add New Institution

18

Total Institutions
All Time

16

Active Institutions
Currently

2

Inactive Institutions
Currently

6

This Month
Newly Added

All Status ▾

All Types ▾

All Locations ▾

↺ Reset

#	Institution Name	Type	Location	Contact Person	Email	Status	Added On	Actions
1	Government College of Engineering	Government	Hyderabad, TS	Dr. Ramesh Kumar	principal@gceh.ac.in	Active	20 May 2024	View Edit Delete
2	University of Technology	University	Warangal, TS	Dr. Anita Devi	admin@uot.ac.in	Active	18 May 2024	View Edit Delete
3	St. Mary's College	Private College	Secunderabad, TS	Mr. John Mathew	info@stmarys.ac.in	Active	15 May 2024	View Edit Delete
4	Vignan Institute of Technology	Private College	Visakhapatnam, AP	Dr. K. Srinivas	principal@vignanit.ac.in	Active	12 May 2024	View Edit Delete
5	CBIT College	Private College	Hyderabad, TS	Dr. P. Reddy	admin@cbit.ac.in	Active	10 May 2024	View Edit Delete
6	Andhra University	University	Visakhapatnam, AP	Prof. M. Rao	registrar@andhrauniv.edu.in	Active	08 May 2024	View Edit Delete
7	Nizam College	Government	Hyderabad, TS	Dr. Syed Ali	principal@nizamcollege.ac.in	Inactive	05 May 2024	View Edit Delete
8	Loyola Academy	Private College	Secunderabad, TS	Fr. Thomas	info@loyolaacademy.ac.in	Active	04 May 2024	View Edit Delete
9	JNTU Hyderabad	University	Hyderabad, TS	Dr. G. Ravinder	registrar@jntuh.ac.in	Active	02 May 2024	View Edit Delete
10	MGIT College	Private College	Hyderabad, TS	Dr. Sujatha	principal@mgit.ac.in	Inactive	01 May 2024	View Edit Delete

Showing 1 to 10 of 18 entries

[◀](#)
1
2
[▶](#)

9. Department Management (UI Wireframe)

**LRUP**
Lab Resource
Utilization Platform

Dashboard

Equipment

Bookings

Maintenance

Calibration

Institutions

Departments

Users

Reports

Notifications

Settings

Logout

Search anything...

Admin User
Administrator

Department Management
Manage departments and their information

+ Add New Department

16
Total Departments
All Time

14
Active Departments
Currently

2
Inactive Departments
Currently

4
This Month
Newly Added

Search departments...

All Status

All Institutions

Reset

#	Department Name	Institution	Head of Department	Email	Status	Created On	Actions		
1	Biotechnology	Government College of Engineering	Dr. Ramesh Kumar	hod.bt@gceh.ac.in	Active	20 May 2024			
2	Chemistry	University of Technology	Dr. Anita Devi	hod.chem@uot.ac.in	Active	18 May 2024			
3	Mechanical Engineering	St. Mary's College	Dr. John Mathew	hod.me@stmarys.ac.in	Active	15 May 2024			
4	Computer Science	Vignan Institute of Technology	Dr. K. Srinivas	hod.cs@vignanit.ac.in	Active	12 May 2024			
5	Electrical Engineering	CBIT College	Dr. P. Reddy	hod.ee@cbit.ac.in	Active	10 May 2024			
6	Microbiology	Andhra University	Prof. M. Rao	hod.micro@andhrauniv.edu.in	Active	08 May 2024			
7	Civil Engineering	Nizam College	Dr. Syed Ali	hod.civil@nizamcollege.ac.in	Inactive	05 May 2024			
8	Physics	Loyola Academy	Fr. Thomas	hod.physics@loyola.ac.in	Active	04 May 2024			
9	Mathematics	JNTU Hyderabad	Dr. G. Ravinder	hod.maths@jntuh.ac.in	Active	02 May 2024			
10	Information Technology	MGIT College	Dr. Sujatha	hod.it@mgit.ac.in	Inactive	01 May 2024			

Showing 1 to 10 of 16 entries

<

1

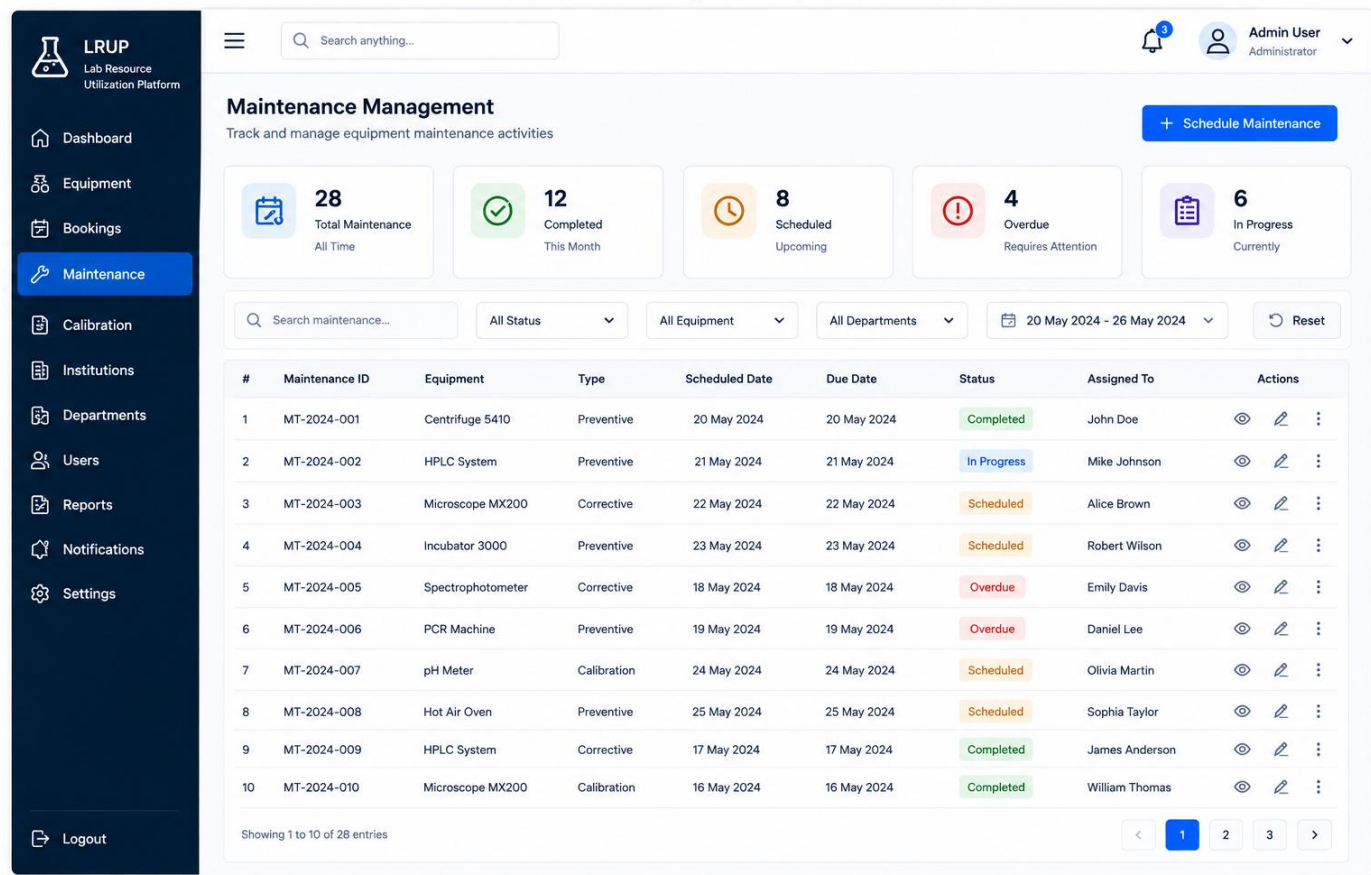
2

>

Administrators can create institutions, manage departments, and assign users and equipment to the respective departments.

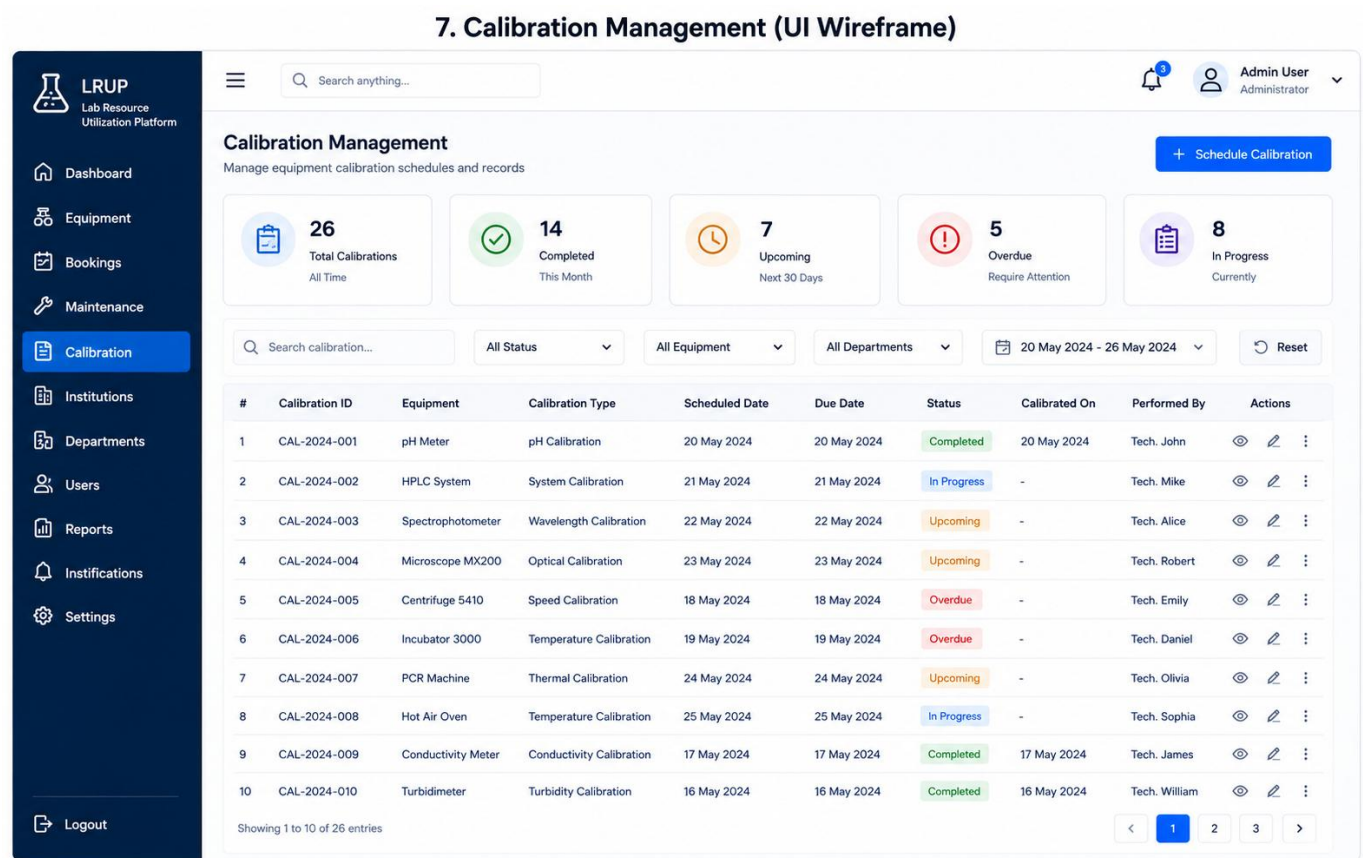
8.8 Maintenance Module

6. Maintenance Management (UI Wireframe)



The Maintenance screen allows administrators to schedule preventive maintenance, assign technicians, and monitor equipment maintenance status.

8.9 Calibration Module



The Calibration module records calibration schedules, certificates, due dates, and equipment calibration history.

User Management:

10. User Management (UI Wireframe)



LRUP
Lab Resource
Utilization Platform

Dashboard

Equipment

Bookings

Maintenance

Calibration

Institutions

Departments

Users

Reports

Notifications

Settings

Logout

Search anything...

48

Total Users

All Time

40

Active Users

Currently

5

Inactive Users

Currently

3

Locked Users

Currently

12

This Month

Newly Added

Search users...

All Roles

All Status

All Institutions

Reset

#	Name	Email	Role	Institution / Department	Status	Last Login	Actions
1	AJ Admin User	admin@lrup.com	Administrator	System	Active	20 May 2024, 10:30 AM	
2	RJ Dr. Ramesh Kumar	ramesh@gceh.ac.in	Head of Department	Govt. College of Engg. / Biotechnology	Active	20 May 2024, 09:15 AM	
3	AD Dr. Anita Devi	anita@uot.ac.in	Department User	University of Technology / Chemistry	Active	19 May 2024, 04:45 PM	
4	JM John Mathew	john@stmarys.ac.in	Department User	St. Mary's College / Mech. Engg.	Active	19 May 2024, 02:20 PM	
5	AS Alice Brown	alice@vignan.it.ac.in	Lab Technician	Vignan Institute / Computer Science	Active	18 May 2024, 11:10 AM	
6	RP Robert Wilson	robert@cbit.ac.in	Lab Technician	CBIT College / Electrical Engg.	Inactive	10 May 2024, 03:30 PM	
7	ED Emily Davis	emily@andhrauniv.edu.in	Student	Andhra University / Microbiology	Active	20 May 2024, 08:50 AM	
8	DL Daniel Lee	daniel@nizamcollege.ac.in	Student	Nizam College / Civil Engg.	Inactive	05 May 2024, 01:15 PM	
9	ST Sophia Taylor	sophia@loyola.ac.in	Student	Loyola Academy / Physics	Active	19 May 2024, 06:00 PM	
10	WT William Thomas	william@mgit.ac.in	Lab Technician	MGIT College / Information Tech.	Locked	02 May 2024, 09:40 AM	

Showing 1 to 10 of 48 entries

<

1

2

3

4

5

>

Notifications:

11. Notification Management (UI Wireframe)



LRUP
Lab Resource
Utilization Platform

Dashboard

Equipment

Bookings

Maintenance

Calibration

Institutions

Departments

Users

Reports

Notifications

Settings

Logout

Search anything...



2



Admin User
Administrator

Notification Management

View and manage all system notifications

+ Send New Notification



126
Total Notifications
All Time



72
Sent
All Time



48
Read
All Time



6
Failed
All Time



24
This Month
Newly Sent

Search notifications...

All Types

All Status

All Recipients

20 May 2024 - 26 May 2024

Reset

#	Title	Type	Message	Recipients	Status	Sent On	Actions
1	Equipment Maintenance Due	Maintenance	Centrifuge 5410 is due for maintenance on 25 May 2024.	Lab Technicians (5)	Sent	20 May 2024, 10:30 AM	  
2	Booking Approved	Booking	Your booking for HPLC System has been approved.	Dr. Ramesh Kumar	Read	20 May 2024, 09:15 AM	  
3	Calibration Reminder	Calibration	pH Meter calibration is scheduled on 22 May 2024.	Lab In-charges (3)	Sent	19 May 2024, 04:45 PM	  
4	New User Registered	System	New user Dr. Anita Devi has been registered.	Administrators (2)	Read	19 May 2024, 02:20 PM	  
5	Equipment Out of Service	Alert	Microscope MX200 is temporarily out of service.	All Lab Users (28)	Sent	18 May 2024, 11:10 AM	  
6	Department Added	System	Department 'Biotechnology' has been added.	Administrators (2)	Read	18 May 2024, 10:05 AM	  
7	Overdue Maintenance	Maintenance	HPLC System maintenance is overdue.	Lab Technicians (5)	Failed	17 May 2024, 03:30 PM	  
8	Booking Cancellation	Booking	Your booking for Incubator 3000 has been cancelled.	Alice Brown	Read	17 May 2024, 01:15 PM	  
9	System Update	System	System will be updated on 21 May 2024 at 10:00 PM.	All Users (48)	Sent	16 May 2024, 05:00 PM	  
10	Calibration Completed	Calibration	Calibration of Spectrophotometer completed.	Lab In-charges (3)	Read	16 May 2024, 11:20 AM	  

Showing 1 to 10 of 126 entries

<

1

2

3

4

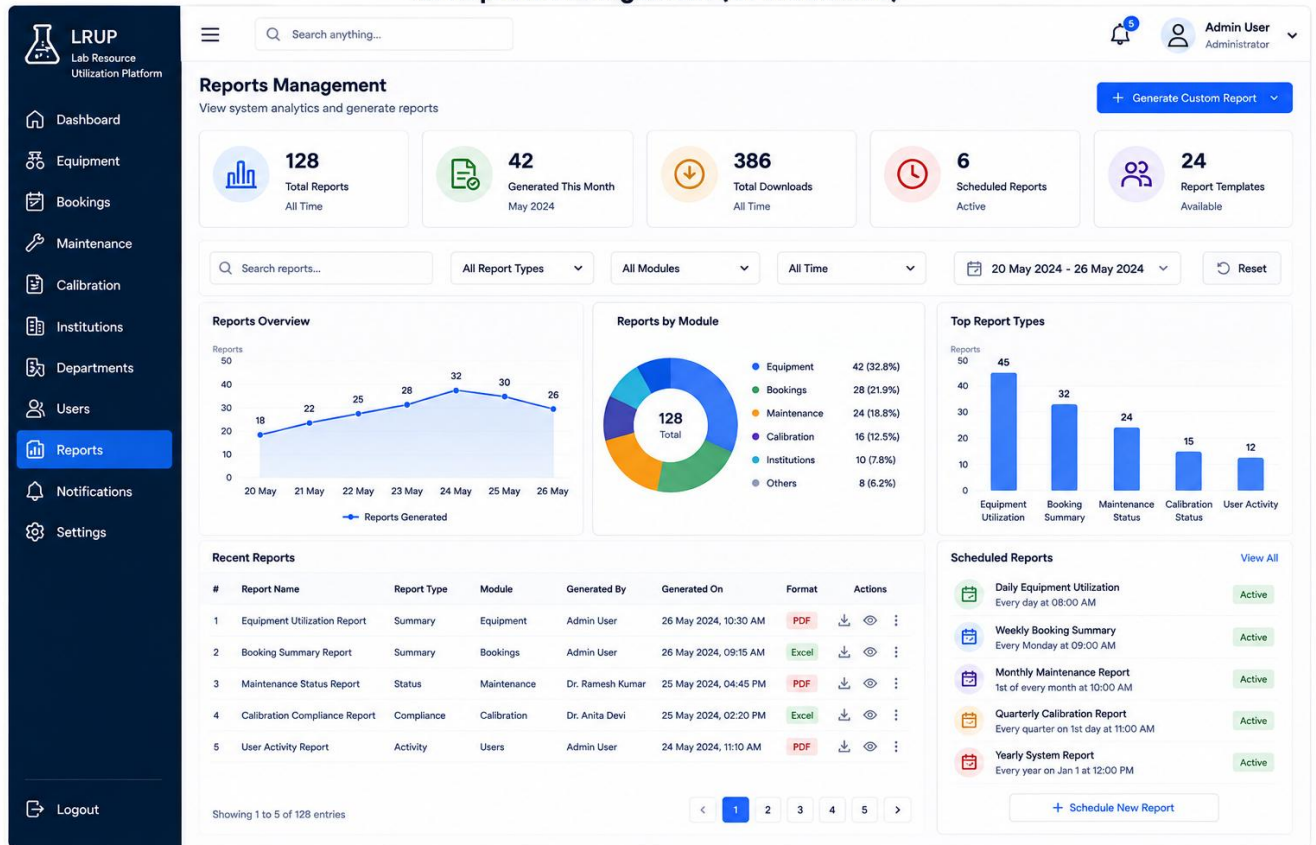
5

>

8.10 Reports Dashboard

The Reports module enables users to generate analytical reports based on equipment utilization, maintenance, bookings, calibration, billing, and operational costs.

12. Reports Management (UI Wireframe)



8.11 Conclusion

The UI wireframes provide a clear visualization of the application's navigation and functionality. They serve as the foundation for frontend implementation and help ensure consistency across all modules of the Lab Resource Utilization Platform.